PRACTICAL NO. 04

PROBLEM STATEMENT: Write a program in solidity to create Student data. Use the following constructs:

- Structures
- Arrays
- Fallback

Deploy this as smart contract on Ethereum and Observe the transaction fee and Gas values.

CODE:

Smart Contract(Solidity)

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.0;

contract StudentData {
    // Structure for student information
    struct Student {
        uint id;
        string name;
        uint age;
        string course;
    }

    // Array to store all student records
    Student[] public students;

    // Event to log student addition
    event StudentAdded(uint id, string name);

    // Function to add a student
    function addStudent(uint _id, string memory _name, uint _age, string memory _course)
public {
        students.push(Student(_id, _name, _age, _course));
        emit StudentAdded(_id, _name);
    }

    // Function to get total students count
    function getStudentCount() public view returns (uint) {
```

```
        return students.length;
    }

    // Fallback function (runs if no function matches)
    fallback() external payable {
        revert("Function not found! Use addStudent() or getStudentCount()");
    }


    // Receive function for plain Ether transfers
    receive() external payable {}
}
```

Output:

